

ACLIM Level 4 Pipeline

A stepwise approach to generate Level 4 bias-corrected ACLIM indices in R

Kirstin Holsman

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1 Overview

This pipeline will step through the methods to create the bias corrected L4 indices for ACLIM. The general approach is:

1. Get the Level3 (and or Level2) files from the thredds ACLIM PMEL server.
2. Convert the Level3 files to Rdata files.
3. Bias correct the projections using the updated hindcast.
4. Make Level4 indices: summarize the weekly strata-specific files by NEBS, SEBS, and NEBS+SEBS area weighted monthly, seasonal and annual means (Level4).
5. Upload the files to the AKFIN database

2 Step 0 : Set up the workspace

```
# rm(list=ls())
# install.packages("here")
library(here) |> suppressMessages() |> suppressWarnings()
main <- here::here()
print(main)

# -----
# load functions and packages in ACLIM repo
# -----
source(here("R/make.R")) |> suppressMessages() |> suppressWarnings()

# -----
# ref_years - for bias correction,
# standardized for ACLIM modelers; update if needed
# -----
ref_years

# -----
# timestamp for updated folder (date is date of download from thredds)
# -----
tmstp    <- "2026_02_05"

# -----
# switches
# -----
# set these to TRUE to update:
redownload_nc_files  <- FALSE
updateR_files        <- FALSE
update_biascorrection <- FALSE

# -----
# set up paths and folders:
# -----
local_fl <- file.path(main,"Data/in",tmstp)
Rdata_path  <- file.path(local_fl,"Rdata_files")
nc_data_path <- file.path(local_fl,"nc_files")
```

```

if(!file.exists(local_fl)) dir.create(local_fl)
if(!file.exists(Rdata_path)) dir.create(Rdata_path)
if(!file.exists(nc_data_path)) dir.create(nc_data_path)

# -----
# load thredds scraping utilities
# -----
source(here("R/subfun_00_aclim_thredds_util.R"))

# -----
# load/get the set of model runs from the thredds server:
# -----

if(redownload_nc_files){
  sims <- aclim_thredds(section = "files")
  save_as(x = sims,
          name = "model_runs",
          file = file.path(Rdata_path, "model_runs.Rdata"))
}else{
  load(file.path(Rdata_path, "model_runs.Rdata")) # model_runs
}

# -----
# load/get the list of Level2 variables from the thredds server:
# -----

if(redownload_nc_files){
  L2_time_periods <- aclim_thredds(run = model_runs[2], level = 2)
  L2_time_periods
  save_as(x = L2_time_periods,
          name = "l2_time_periods",
          file = file.path(Rdata_path, "l2_time_periods.Rdata") )

  L2_files <- aclim_thredds(run = model_runs[2],
                           level = 2,
                           time_period = L2_time_periods[1] )

```

```

L2_var_list <- sub("^average_", "", L2_files$variable)
save_as(L2_var_list,
        name = "l2_vars",
        file = file.path(Rdata_path, "l2_vars.Rdata") )

}else{
  load(file.path(Rdata_path, "l2_vars.Rdata"))
}

```

3 Step 1 & 2: get the Level3 files from thredds

The code below uses the `get_l3` function - updated in 2026 - to grab the Level3 (L3) .nc files from the ACLIM thredds PMEL server and convert them to Rdata files (locally). This is focused on the L3 files; to download the .nc files for the 2D gridded (L2) and/or L3 directly see: `~ACLIM2/Vignettes/get_ACLIM_nc_thredds_data.R` which includes step by step methods to inventory and download both Level2 and Level3 indices.

```

# =====
# grab the .nc files from the thredds server:
# =====
if(redownload_nc_files){

  get_l3(from_thredds = T,
        download_nc = T,
        cleanIT = F,
        convert2Rdata = F, # just get the data for now
        local_path = NULL,
        varlist = vl,
        sim_list = model_runs[-which(model_runs%in%c("B10K-K20nobio_CORECFS_daily"))])
  # get all the data

}

# =====
# now convert the nc files to rdata for the variables in vl
# =====
tmp_path <- here::here("Data/in/2026_02_05/nc_files/B10K-K20_CORECFS/Level3")

nc <- nc_open(file.path(tmp_path, "ACLIMsurveyrep_B10K-K20_CORECFS.nc"))

```

```

survey_vars <- names(nc$var)
nc_close(nc)

nc <- nc_open(file.path(tmp_path,"ACLIMregion_B10K-K20_CORECFS.nc"))
weeklyByarea_vars <- names(nc$var)
nc_close(nc)
# weekly_vars may be outdated

setdiff(weeklyByarea_vars,weekly_vars)

# -----
# optional: first test with a small set of vars such as:
# -----
v1 <- c("temp_bottom5m" ,
        "NCaS_integrated",
        "Cop_integrated" ,
        "EupS_integrated")

# weekly_vars and srvy_vars are lists loaded in the R/make.R from the ACLIM2 repo
all_vars <- c(
  weeklyByarea_vars[-which(weeklyByarea_vars%in%c("region_area"))],
  survey_vars[-which(survey_vars%in%c("station_id",
                                       "latitude",
                                       "longitude",
                                       "stratum",
                                       "doy"))]) |>unique()

skip_list <-c("Iron_bottom5m","Iron_integrated","Iron_surface5m")
save(skip_list,file=file.path(Rdata_path,"skip_list.Rdata"))

# this takes a while if downloading all of the vars (approx 5 min per model run) ....
if(updateR_files){

  get_l3(from_thredds = F,
        download_nc = F,
        cleanIT = T,
        convert2Rdata = T,
        fileset = c("ACLIMsurveyrep", "ACLIMregion" ),
        varlist = all_vars[!all_vars%in%skip_list],

```

```

    local_path = NULL,
    verbose = T,
    sim_list = model_runs[!model_runs%in%
                          c("B10K-K20nobio_CORECFS_daily")]])

# get all the data

}

```

4 Step 3: Bias correct the projections.

In the next step we bias correct the projections based on historical or hindcast data. For CMIP5, we did not have historical runs for each global model (e.g., MIROC) so we bias corrected to a reference period from the hindcast (re-centering each variable in each projection using the overlapping projection and hindcast periods such that the mean of the projection was adjusted downward (upward) to match the mean of the corresponding hindcast reference period). This resulted in a slightly smaller “overlapping” reference period.

For CMIP6, we had historical time periods that extended further back for each projection simulation so we recentered the projection x global model combination (e.g., MIROC ssp126) on the delta adjusted historical reference period means (historical MIROC and hindcast).

For survey data the bias correction occurs at the level of the strata (not station location).

For the regional weekly data, bias correction occurred at the strata and weekly using a gam smoothed climatological average (to interpolate / smooth means across the few times when model weeks were missing from the simulation year).

These methods are explored in more detail in Bigmann et al, in review.

```

# =====
# now bias correct to the hindcast for each CMIP
# =====

# define the CMIP model sets:
CMIP6_set <- model_runs[grep("K20P19",model_runs)]
CMIP5_H16_set <- model_runs[grep("H16_",model_runs)]

# define the hindcast (should only be one for each)
hind_cmip5_H16_nm <- CMIP5_set[grep("CORECFS",CMIP5_set)]
#hind_cmip5_K20P19_nm <- CMIP5_set[grep("CORECFS",CMIP5_set)]
hind_cmip6_nm <- CMIP6_set[grep("CORECFS",CMIP6_set)]

```

```

# -----
# load the weekly_vars list
# -----
if(updateR_files){
  L3_types <- c("ACLIMsurveyrep_", "ACLIMregion_")

  tmppath <- file.path(nc_data_path, CMIP6_set[d], "Level3")
  dirlist <- dir(tmppath)
  tmp_fl <- paste0(L3_types[1], CMIP6_set[d], ".nc")

  nc <- ncdf4::nc_open(file.path(tmppath, tmp_fl))
  print(nc)

  srvy_vars <- names(nc$var)
  srvy_vars <- c(srvy_vars, "largeZoop_integrated")

  save(srvy_vars, file = file.path(Rdata_path, "srvy_vars.Rdata"))
  nc_close(nc)

  tmppath <- file.path(nc_data_path, CMIP6_set[d], "Level3")
  dirlist <- dir(tmppath)
  tmp_fl <- paste0(L3_types[2], CMIP6_set[d], ".nc")

  nc <- ncdf4::nc_open(file.path(tmppath, tmp_fl))
  print(nc)

  weekly_vars <- names(nc$var)
  weekly_vars <- c(weekly_vars, "largeZoop_integrated")
  save(weekly_vars, file = file.path(Rdata_path, "weekly_vars.Rdata"))
  nc_close(nc)

}else{
  load(file.path(Rdata_path, "weekly_vars.Rdata"))
  load(file.path(Rdata_path, "srvy_vars.Rdata"))
  load(file.path(Rdata_path, "l2_vars.Rdata"))
  load(file.path(Rdata_path, "skip_list.Rdata"))
}

```

```

# -----
# now create the bias corrected L4 Indices
# -----
# skip_list

vl1  <- srvy_vars #srvy_vars[!srvy_vars%in%rm_var_list]
vl2  <- weekly_vars # weekly_vars[!weekly_vars%in%rm_wk_list]

# -----
# add in largeZoop (gets generated in make_indices_region_new.R)
# -----
vl <- c(unique(c(vl1,vl2)),"largeZoop_integrated")

# -----
# Identify which variables would be normally
# distributed (i.e., can have negative values)
# -----
update_norm_list <- FALSE
if(update_norm_list){

  normvl <- c(
    vl[grepl("temp",vl)],
    vl[grepl("pH",vl)],
    #vl[grepl("Ben",vl)],
    vl[grepl("Hsbl",vl)],
    vl[grepl("shflux",vl)],
    vl[grepl("ssflux",vl)],
    vl[grepl("vNorth",vl)],
    vl[grepl("uEast",vl)])

  normlist <- data.frame(var = vl, lognorm = "none",stringsAsFactors = F)
  normlist$lognorm[!normlist$var%in%normvl] <- "log"

  # specify which can only be 0 to 1 as "logit"
  normlist$lognorm[normlist$var%in%
    c( vl[grepl("aice",vl)],
      vl[grepl("fracbelow0",vl)],
      vl[grepl("fracbelow1",vl)],
      vl[grepl("fracbelow2",vl)])] <- "logit"

```



```

save(normlist,file      = file.path(Rdata_path,"normlist.Rdata"))
write.csv(normlist,file = file.path(Rdata_path,"normlist.csv"))

}else{
  load(file      = file.path(Rdata_path,"normlist.Rdata"))
}

sv_bc      <- c("largeZoop_integrated","fracbelow2",
               "temp_bottom5m","temp_surface5m","pH_depthavg")
sv_bc <- NULL # bias correct all indices

save(weekly_vars,file  = here::here("Data/out/weekly_vars.Rdata"))
save(srvy_vars,file    = here::here("Data/out/srvy_vars.Rdata"))
write.csv(normlist,file = here::here("Data/out/", "normlist.csv"))

# =====
# generate indices and bias corrected projections
# this takes about 30 mins each
# =====

load(file.path(Rdata_path,"model_runs.Rdata"))

gcmcmip6 <- c("B10K-K20P19_CMIP6_miroc",
             "B10K-K20P19_CMIP6_gfdl",
             "B10K-K20P19_CMIP6_cesm")

#source("~/Documents/GitHub_mac/ACLIM2/R/03_load_functions.R")
rmlist <- c("station_id","latitude","longitude","stratum","doy")

# -----
# =====
# CMIP6 K20P19
# =====

```

```

ref_yearsSET <- 1980:2013
# The problematic variables were prod_PhL_integrated and
# prod_PhS_integrated and the NAs were all of 2020 and 2021 for both.

# slow....
# -----
# weekly: first run hind, hist, and projections for strata-weekly:
# -----
tmp_hind <-
  makeACLIM2_BC_Indices_new(
    bystrata = TRUE, #<----- KEY# if T run indices on regnm; F run indices for srvyrm
    BC_target = "mn_val", # bias corrected column name
    ref_yrsIN = ref_yearsSET,
    sv = sv_bc,
    updateHist = TRUE, # update_hist,
    updateHind = TRUE, # update_hind,
    updateProj = TRUE,
    smoothIT = TRUE, # use gam() smoother
    sfIN = "val_delta",
    CMIP_fdlr = "K20P19_CMIP6",
    scenIN = c("ssp126", "ssp585"),
    hind_sim = "B10K-K20P19_CORECFS",
    histLIST = paste0(gcmcmip6, "_historical"),
    gcmcmipLIST = gcmcmip6,
    usehist = TRUE,
    Rdata_pathIN = Rdata_path,
    regnm = "ACLIMregion",
    srvyrm = "ACLIMsurveyrep",
    normlist_IN = normlist[!normlist$var%in%rmlist,],
    sim_listIN = model_runs[!grep("historical", model_runs)],
    gcinfoIN = FALSE) |> suppressMessages() |> suppressWarnings()

#rplc_bcplot <-TRUE # update the bias corrected plot
#source(here("R/sub_scripts/plot_BC_stratawk.R"))
# check prod_PhL_integrated and prod_PhS_integrated between 2020 and 2021
# looks good!

# -----
# survey_replicated: now skip hind & hist and run projections for survey replicated:

```

```

# -----
tmp2_hind<-
  makeACLIM2_BC_Indices_new(
    bystrata = TRUE,          #<----- KEY
    BC_target = "mn_val",
    sv       = sv_bc,
    sfIN     = "val_delta",
    smoothIT = TRUE, # use gam() smoother
    updateHist = TRUE, # update_hist,
    updateHind = TRUE, # update_hind,
    ref_yrsIN = ref_yearsSET,
    updateProj = TRUE,
    CMIP_fdlr  = "K20P19_CMIP6",
    scenIN     = c("ssp126", "ssp585"),
    hind_sim   = "B10K-K20P19_CORECFS",
    histLIST   = paste0(gcmcmip6, "_historical"),
    gcmcmipLIST = gcmcmip6,
    usehist    = TRUE,
    Rdata_pathIN = Rdata_path,
    regnm      = "ACLIMregion",
    srvynm     = "ACLIMsurveyrep",
    normlist_IN = normlist[!normlist$var%in%rmlist,],
    sim_listIN  = model_runs[-grep("historical", model_runs)],
    gcinfoIN    = FALSE) |> suppressMessages() |> suppressWarnings()

# =====
# CMIP5 K20P19
# =====
ref_yearsSET <- 1980:2020
gcmcmip5_K20P19 <- c( "B10K-K20P19_CMIP5_MIROC",
                      "B10K-K20P19_CMIP5_GFDL",
                      "B10K-K20P19_CMIP5_CESM")

# -----
# weekly: make bc indices for hindcast and strata-weekly:
# -----
tmphind <- makeACLIM2_BC_Indices_new(
  bystrata = TRUE,          #<----- KEY
  BC_target = "mn_val",

```

```

ref_yrsIN = ref_yearsSET,
sv        = sv_bc,
updateHist = FALSE, # update_hist,
updateHind = TRUE, # update_hind,
updateProj = TRUE,
smoothIT   = TRUE,
CMIP_fdlr  = "K20P19_CMIP5",
scenIN     = c("rcp45", "rcp85"),
hind_sim   = "B10K-K20P19_CORECFS",
histLIST   = gcmcmip5_K20P19,
gcmcmipLIST = gcmcmip5_K20P19,
usehist     = FALSE,
Rdata_pathIN = Rdata_path,
regnm       = "ACLIMregion",
srvynm      = "ACLIMsurveyrep",
normlist_IN = normlist[!normlist$var%in%rmlist,],
sim_listIN  = model_runs[-grep("historical", model_runs)],
gcinfoIN    = FALSE) |> suppressMessages() |> suppressWarnings()

# -----
# survey replicated: now skip hind & hist and run projections for survey replicated:
# -----

tmp <- makeACLIM2_BC_Indices_new(
  bystrata = FALSE, #<----- KEY
  BC_target = "mn_val",
  ref_yrsIN = ref_yearsSET,
  sv        = sv_bc,
  updateHist = FALSE, # update_hist,
  updateHind = TRUE, # update_hind,
  updateProj = TRUE,
  smoothIT   = TRUE,
  CMIP_fdlr  = "K20P19_CMIP5",
  scenIN     = c("rcp45", "rcp85"),
  hind_sim   = "B10K-K20P19_CORECFS",
  histLIST   = gcmcmip5_K20P19,
  gcmcmipLIST = gcmcmip5_K20P19,
  usehist     = FALSE,
  Rdata_pathIN = Rdata_path,

```

```

regnm      = "ACLIMregion",
srvynm     = "ACLIMsurveyrep",
normlist_IN = normlist[!normlist$var%in%rmlist,],
sim_listIN  = model_runs[-grep("historical",model_runs)],
gcinfoIN    = FALSE) |> suppressMessages() |> suppressWarnings()

# rplc_bcplot <-TRUE
# source("R/sub_scripts/plot_BC_stratawk.R")

# =====
# CMIP5 H16 (older version of the ROMSNPZ model without OA dynamics )
# =====

ref_yrsSET <- 1980:2016

gcmcmip5_H16 <- c("B10K-H16_CMIP5_MIROC",
                 "B10K-H16_CMIP5_GFDL",
                 "B10K-H16_CMIP5_CESM",
                 "B10K-H16_CMIP5_CESM_BIO")

# -----
# weekly: now run hind and run projections for weekly-strata:
# -----

tmphind <- makeACLIM2_BC_Indices_new(
  bystrata = TRUE,                #<----- KEY
  BC_target = "mn_val",
  sv        = sv_bc,
  ref_yrsIN = ref_yrsSET,
  updateHist = FALSE, # update_hist,
  updateHind = TRUE,  # update_hind,
  updateProj = TRUE,
  smoothIT   = TRUE,
  CMIP_fdlr  ="H16_CMIP5",
  scenIN     = c("rcp45","rcp85"),
  hind_sim   = "B10K-H16_CORECFS",
  histLIST   = gcmcmip5_H16,
  gcmcmipLIST = gcmcmip5_H16,
  usehist    = FALSE,
  Rdata_pathIN = Rdata_path,

```

```

regnm      = "ACLIMregion",
srvynm     = "ACLIMsurveyrep",
normlist_IN = normlist[!normlist$var%in%rmlist,],
sim_listIN  = model_runs[-grep("historical",model_runs)],
gcinfoIN    = FALSE) |> suppressMessages() |> suppressWarnings()

# -----
# survey_replicated: now skip hind & and run projections for survey replicated:
# -----

tmp <- makeACLIM2_BC_Indices_new(
  bystrata = FALSE,          #<----- KEY
  BC_target = "mn_val",
  ref_yrsIN = ref_yrsSET,
  sv        = sv_bc,
  updateHist = FALSE, # update_hist,
  updateHind = FALSE, # update_hind,
  updateProj = TRUE,
  smoothIT   = TRUE,
  CMIP_fdlr  = "H16_CMIP5",
  scenIN     = c("rcp45","rcp85"),
  hind_sim   = "B10K-H16_CORECFS",
  histLIST   = gcmcmip5_H16,
  gcmcmipLIST = gcmcmip5_H16,
  usehist    = FALSE,
  Rdata_pathIN = Rdata_path,
  regnm      = "ACLIMregion",
  srvynm     = "ACLIMsurveyrep",
  normlist_IN = normlist[!normlist$var%in%rmlist,],
  sim_listIN  = model_runs[-grep("historical",model_runs)],
  gcinfoIN    = FALSE) |> suppressMessages() |> suppressWarnings()

# rplc_bcplot <-TRUE
# source("R/sub_scripts/plot_BC_strataawk.R")
#

```

5 Step 4: Make Level4 ACLIM indices

In this step we summarize the weekly strata-specific files by NEBS, SEBS, and NEBS+SEBS area weighted weekly, monthly, seasonal and annual means as well as survey replicated annual indices (Level4).

```
# make ACLIM level 4 indices (annual, monthly, weekly, etc)
# source("~/Documents/GitHub_mac/ACLIM2/R/03_load_functions.R")

# =====
# CMIP6
# =====

suppressMessages(
  makeACLIM2_L4_Indices_strata(
    CMIP_fdlr = here::here("Data/out/K20P19_CMIP6"),
    CMIP      = "CMIP6",
    scenIN    = c("ssp126", "ssp585"),
    hind_sim  = "B10K-K20P19_CORECFS",
    gcmcmipLIST = gcmcmip6,
    ref_yrsIN = 1980:2013,
    subfldrIN = "BC_ACLIMregion",
    sim_listIN = model_runs,
    varlistIN = sv_bc,
    prefix    = "ACLIMregion"))

suppressMessages(makeACLIM2_L4_Indices_survey(
  CMIP_fdlr = here::here("Data/out/K20P19_CMIP6"),
  CMIP      = "CMIP6",
  usehist   = TRUE,
  scenIN    = c("ssp126", "ssp585"),
  hind_sim  = "B10K-K20P19_CORECFS",
  gcmcmipLIST = gcmcmip6,
  subfldrIN = "BC_ACLIMsurveyrep",
  sim_listIN = model_runs,
  varlistIN = sv_bc,
  prefix    = "ACLIMsurveyrep"))

source(here::here("R/sub_scripts/plot_NEBSnSEBS.R"))
```

```

# =====
# CMIP5 - K20P19
# =====

suppressMessages(makeACLIM2_L4_Indices_strata(
  CMIP_fdlr = here::here("Data/out/K20P19_CMIP5"),
  CMIP      = "CMIP5",
  scenIN    = c("rcp45","rcp85"),
  hind_sim  = "B10K-K20P19_CORECFS",
  gcmcmipLIST = gcmcmip5_K20P19,
  ref_yrsIN = 1980:2020,
  subfldrIN = "BC_ACLIMregion",
  sim_listIN = model_runs,
  varlistIN  = sv_bc,
  prefix     = "ACLIMregion"))
gc()

suppressMessages(makeACLIM2_L4_Indices_survey(
  CMIP_fdlr = here::here("Data/out/K20P19_CMIP5"),
  CMIP      = "CMIP5",
  usehist   = FALSE,
  scenIN    = c("rcp45","rcp85"),
  hind_sim  = "B10K-K20P19_CORECFS",
  gcmcmipLIST = gcmcmip5_K20P19,
  subfldrIN = "BC_ACLIMsurveyrep",
  sim_listIN = model_runs,
  varlistIN  = sv_bc,
  prefix     = "ACLIMsurveyrep"))

# =====
# CMIP5 H16
# =====

suppressMessages(makeACLIM2_L4_Indices_strata(
  CMIP_fdlr = here::here("Data/out/H16_CMIP5"),
  CMIP      = "CMIP5",
  scenIN    = c("rcp45","rcp85"),
  hind_sim  = "B10K-H16_CORECFS",
  gcmcmipLIST = gcmcmip5_H16,
  ref_yrsIN = 1980:2016,

```



```

subfldrIN  = "BC_ACLIMregion",
sim_listIN = model_runs,
varlistIN  = sv_bc,
prefix     = "ACLIMregion"))
gc()

suppressMessages(makeACLIM2_L4_Indices_survey(
  CMIP_fldr  = here::here("Data/out/H16_CMIP5"),
  usehist    = FALSE,
  CMIP       = "CMIP5",
  scenIN     = c("rcp45","rcp85"),
  hind_sim   = "B10K-H16_CORECFS",
  gcmcmipLIST = gcmcmip5_H16,
  subfldrIN  = "BC_ACLIMsurveyrep",
  sim_listIN = model_runs,
  varlistIN  = sv_bc,
  prefix     = "ACLIMsurveyrep"))

```

6 Step 5: Upload the files to the AKFIN database

7 APP A: Misc older code - ignore